

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)
ON
SKILLGAP AI: A SMART SKILLING ASSESSMENT AND INDIVIDUALIZED
COURSE RECOMMENDATION SYSTEM

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**SKILLGAP AI: A SMART SKILLING ASSESSMENT AND INDIVIDUALIZED COURSE RECOMMENDATION SYSTEM**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Swati Goel. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

The SkillGap AI project is called SkillGap AI: A Smart Skilling Assessment and Individualized Course Recommendation System. This project is trying to find the difference between the skills a person has now and the skills they need for a job.

The SkillGap AI system uses machine learning to suggest career options for people based on their skills. It looks at what people write and turns it into numbers so it can be understood better. The system uses different methods to make sure its predictions are correct.

The SkillGap AI system also looks at what skills a person already has and suggests training or courses that can help them get the skills they're missing. This helps people focus on getting the skills they need for the job they want.

The SkillGap AI system is made to help people get the skills and make good choices about their career. The SkillGap AI system is really good at guiding people to develop the skills they need. The SkillGap AI project is, about the SkillGap AI system.

Introduction

In today's world having the right skills is crucial for growing in your career. Many people do not know what skills are needed for jobs. This leads to a gap, between what they can do and what employers want.

The SkillGap AI system helps solve this problem. It looks at the skills you have. Suggests career paths that fit you. It also finds out what skills you are missing. Recommends courses to help you get better.

This project uses machine learning and a recommendation system to provide practical career guidance.

Methodology

1. Data Collection

The system uses a career dataset that has information about skills and the career options that go with them. This career dataset is used to train machine learning models so they can learn from the data.

2. Data Preprocessing

The data we collect needs to be cleaned up. Made ready before we can use it to train machine learning models. Here are the things we do:

- We make all the text lowercase
- We remove any entries that're the same
- We make sure the labels for skills and careers are all the same

This step helps make sure everything is consistent and it makes the models work better.

3. Feature Extraction

The data we have is in words so we use something called TF-IDF to turn it into numbers that the model can understand. This helps the model see which words are important in the career dataset.

4. Model Training

We use different machine learning models to figure out which career options are best for the users skills. These models include:

- Decision Tree
- Logistic Regression
- Naïve Bayes
- Random Forest
- SGD Classifier
- K-Nearest Neighbour
- Ridge Classifier

Each of these models is trained using the data we cleaned up.

5. Model Evaluation

We look at how each model does using these measures:

- How often it is right
- How precise it is
- How well it remembers
- The F1 Score

These measures help us compare the models and pick the one.

6. Skill Gap Analysis and Recommendation

After we think we have found a career, for the user we compare the skills they have now with the skills they need for that career. We find the skills they're missing which is called the skill gap. Then we suggest courses that can help the user learn the skills they are missing so they can get the career they want.

7. System Workflow

This is the order we follow:

User Input → Data Preprocessing → Feature Extraction → Model Prediction → Skill Gap Identification → Course Recommendation → Output

The career dataset and machine learning models work together to help the user find a career and learn the skills they need. The system uses the career dataset to train the machine learning models and the machine learning models use the career dataset to make predictions.

Tools and Technologies

Programming Language: Python

Libraries Used : Pandas, NumPy, Matplotlib, Scikit-learn

Machine Learning Techniques: Classification Algorithms

Text Processing: TF-IDF Vectorization

Platform: Google Collab

Implementation

Initially, the program loads and prepares the dataset and transforms any non-numeric values in the text into numerical format via the TF-IDF method.

Once the above-mentioned three steps have been completed, the dataset is divided into 2 sets, i.e., training and testing. The next step is to train a variety of machine learning algorithms on the dataset and then use them to classify jobs according to user skill.

After each algorithm has been trained, they are all evaluated using various performance measures such as accuracy, precision, recall, and F1 score. Also, a skill gap analysis module is created to compare user skills against those required for a job and also identify the skills that need improvement.

Lastly, a recommended course will be provided to the user based on the skills that need to be improved and filled in.

Major Findings/Results

According to the analysis, there were three types of classifiers that significantly outperformed others

The three best performing classifiers with respect to text classification were: Logistic Regression, SGD Classifier, and Ridge Classifier.

The use of TF-IDF created a much more effective representation of the features.

This system has been successful in predicting the user's possible career path.

The skill gap analysis tool was able to accurately identify which skills should be developed.

The course recommendation feature has produced many accurate recommendations.

Conclusion and Future Scope

The SkillGap AI engine analyzes the user's skill set and predicting different career paths along with recommended courses that will help to close any skill gaps identified.

This system offers a useful means for both students and professionals to develop their skills while making more informed choices about careers.

Future Potential

- Utilize larger datasets to improve accuracy
- Utilize deep learning models
- Integrate real-time data from the job market
- Develop either a web or a mobile application